

Alvaro Perdomo Gutierrez

Senior Full Stack Engineer

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with **12+** years across fintech, e-commerce, and healthcare, building scalable platforms in Java and React with cloud infrastructure and strong observability. Experienced in distributed systems, real-time data processing, and production-grade feature delivery using Quarkus, PostgreSQL, and TypeScript with comprehensive testing and CI/CD automation.

SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, HTML, CSS
Backend: Quarkus, Spring Boot, RESTEasy Reactive, CDI, Node.js, Express, FastAPI
Frontend: React, Next.js, Redux, MUI, Tailwind CSS, Webpack, Vite
Database: PostgreSQL, Flyway, Liquibase, Redis, MongoDB, Elasticsearch
Testing: Jest, Pytest, k6, Cypress, JUnit, Mockito, Testcontainers
Cloud & DevOps: Docker, Kubernetes, GitHub Actions, AWS, GCP, Terraform, Jenkins
Others: Kafka, Redpanda, RESTful API, GraphQL, JWT, MicroProfile, Hibernate ORM, Panache, Git, Jira, Confluence, Agile

EXPERIENCE

- LeanwareMay 2024–PresentSenior Full Stack Engineer
- Led the development of an AI-powered algorithm trading platform using **Java**, **Quarkus**, and **React**, delivering end-to-end features from backend microservices to real-time trading dashboards that processed market data across 15+ asset classes.
 - Built and optimized **RESTful APIs** using **Quarkus** with **CDI** dependency injection and **RESTEasy Reactive** to handle high-frequency trading requests, reducing API response latency from 850ms to **280ms** under sustained load.
 - Architected the database layer using **PostgreSQL** with **Flyway** migrations, designing normalized schemas for trade execution and portfolio tracking that supported 50k+ concurrent users without performance degradation.
 - Implemented comprehensive test coverage across **Jest** unit tests and **k6** performance tests, achieving 87% code coverage while automated tests caught 12+ critical integration bugs before production deployment.
 - Extended event-driven order processing using **Kafka** topics to decouple trade execution, settlement, and reporting services, which improved system resilience and cut recovery time from incidents by **65%**.

- Developed the React frontend using **TypeScript**, **Next.js**, and **MUI** components to visualize trading positions, real-time P&L, and compliance alerts, integrating charting libraries to render 10,000+ data points without frame drops.
- Deployed microservices using **Docker** containers orchestrated by **Kubernetes**, configured **GitHub Actions** CI/CD pipelines for automated testing and staging validation, reducing deployment time from 45 minutes to under **6 minutes**.
- Collaborated with the data science team to integrate ML-powered price prediction models into trading recommendations, securing API endpoints with **JWT** token validation and role-based access control for compliance requirements.
- Mentored two junior engineers on full-stack feature ownership, code review discipline, and database migration safety, raising the team's test coverage standards and reducing production incidents by 40%.
- Troubleshooted production incidents in the trading engine, identifying a deadlock in the order queue that had been silently dropping trades under peak load, resolved by adding retry logic and circuit breaker patterns.
- Implemented customer-specific features for institutional traders including custom reporting exports, webhook-based notifications, and role-based portfolio visibility using **PostgreSQL** views and **GraphQL** subscriptions.
- Evaluated and integrated **Redpanda** as a Kafka-compatible message broker, completing migration from self-managed Kafka cluster which reduced operational overhead by 30% and improved cluster stability.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Spearheaded the backend architecture for the AI Shopping Image Search feature, building **Java** microservices with **Spring Boot** to index and search 172M+ resale listings using semantic embeddings and visual features.
- Designed and implemented the search ranking API using **Quarkus**, **RESTEasy Reactive**, and **PostgreSQL** full-text search, optimizing query performance from 2.3 seconds to **310ms** average response time by adding composite indexes and result caching with **Redis**.
- Built the React frontend for Style Chat using **TypeScript**, **Next.js**, and real-time streaming to deliver AI-powered fashion recommendations, achieving a **+2.8%** iOS conversion lift and **+15%** incremental sell-through within the first quarter.
- Migrated the ML feature platform to **Databricks**, coordinating with data science to refactor model deployment pipelines, cutting garment-measurement model iteration from 2 quarters to **1 month** and enabling faster AB testing.
- Developed comprehensive integration tests using **Testcontainers** and **Docker** to validate data pipelines processing 100K+ product images daily, catching schema incompatibilities and race conditions before they reached production.
- Implemented **GitHub Actions** workflows for automated testing and deployment, including performance benchmarks that tracked model inference latency across staging and production environments.
- Extended the pricing recommendation engine using **PostgreSQL** stored procedures and materialized views, which powered dynamic pricing for 55K+ brands and improved sell-through accuracy by 8%.

- Collaborated with product on the Lily AI integration that tagged 275K+ images per week, building scalable backend endpoints to ingest AI predictions and validate data quality, reducing manual tagging backlog by 60%.
- Debugged and resolved a production outage in the auto-photography pipeline that had been incorrectly labeling 5% of garment images, fixed by adding data validation logic and monitoring alerts to **Datadog**.
- Mentored a mid-level engineer on full-stack backend patterns, code review rigor, and database optimization techniques, including lessons learned from scaling feature development from MVP timelines of 2 weeks down from 2 months.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Played a key role in scaling the precision-matching ML platform from 300K to 2M+ members across healthcare plans, building backend services in **Java** and **Spring Boot** that maintained clinical validity while supporting 6x user growth.
- Designed the Care Navigation platform database schema using **PostgreSQL** with complex join logic and views to match patients with appropriate care resources, which was deployed across Fortune 500 clients (PepsiCo, Instacart, Bain).
- Implemented the global family launch feature using **Next.js** and **React**, adding dependent eligibility logic, multi-language localization, and jurisdiction-aware member matching for 18 countries across EMEA and APAC.
- Built operational ML services using **Python** and **FastAPI** to predict member dropout risk and forecast demand for mental health services, integrating predictions into the care recommendation engine via **RESTful API** endpoints.
- Engineered HIPAA-compliant infrastructure scaling on **AWS** to onboard multi-tenant healthcare organizations, implementing field-level encryption, audit logging, and role-based access controls that passed SOC 2 Type II certification.
- Developed integration tests using **Pytest** and Docker containers to validate healthcare data workflows across member enrollment, clinical assessment, and care coordination modules.
- Optimized **PostgreSQL** queries that had been timing out during high-volume member uploads, adding composite indexes and refactoring N+1 queries, reducing upload processing time from 18 minutes to **3 minutes**.
- Collaborated with compliance and security teams to implement JWT-based authentication and API rate limiting, protecting member data while maintaining performance under peak traffic during open enrollment periods.
- Wrote documentation and ADRs (Architecture Decision Records) on feature deployments and schema migrations, establishing best practices for database safety and gradual rollout strategies.

Tencent Games

Nov 2014–Nov 2020

Senior Game Engineer

- Built backend services in **Java** for Honor of Kings, a mobile MOBA that scaled to 100M+ daily active users, implementing match-making, real-time player progression, and leaderboard systems using **PostgreSQL** and distributed caching.
- Designed and maintained the gaming infrastructure using **GCloud** platform and TcaplusDB (a NoSQL store), processing 40M+ daily requests while maintaining sub-100ms latency for match results and player state synchronization.
- Implemented the ACE kernel-level anti-cheat system that detected and prevented unauthorized game modifications, reducing fraudulent accounts by 94% and protecting the competitive integrity of esports tournaments.
- Optimized matchmaking algorithms using **Redis** to cache player skill ratings and availability, reducing queue wait times from 45 seconds to under **8 seconds** during peak concurrent player loads of 5M+.
- Collaborated on cross-platform deployment of Arena of Valor using **Docker** containers, coordinating release engineering for iOS, Android, and console platforms simultaneously across 15 regions.
- Built real-time communication services using event-driven architecture to broadcast match events to spectators and tournament commentators, processing 2M+ concurrent WebSocket connections using custom networking protocols.
- Debugged a critical memory leak in the player session cache that had been causing server crashes during maintenance windows, traced the issue to unbounded GC pressure and resolved by implementing session eviction policies.
- Mentored game engineers on distributed systems design patterns, teaching concepts around eventual consistency, leader election, and failover strategies relevant to managing millions of concurrent player sessions.
- Participated in the FIFA Online 3 operations team, managing live game service stability and responding to player-impacting issues within SLA requirements, maintaining 99.8% uptime through the World Cup esports season.

EDUCATION

The Hong Kong Polytechnic University

Sep 2010–Nov 2014

Bachelor of Science in Computer Science